Student Name					
Teacher	AHU	COZ	GSL	HAL	MDI

CHEMISTRY
Unit 4, SAC 2
Covering concepts from Chapters 15.1 – 15.6 & 16 of Heinemann Chemistry 2
Food Chemistry

Week 10, Friday 17th September

Writing time: 45 minutes

Structure of book

Number of questions	Number of questions to be answered	Number of marks
5	5	27

- Students are permitted to bring into the assessment session:
  - Laptop computers, Data Book (electronic or printed), a scientific calculator

Materials supplied
- Question and answer book of 7 pages by assignments on Teams.
- Data Book

Instructions
- Write your **name** and highlight your **teacher code** in the space provided above on this page.
- All written responses must be in English.
- Answer all questions in the spaces provided.

At the end of the assessment
- Ensure you have saved your work in Word and "Turned in" your task on Microsoft teams.
- Do not close the Word document until confirming that the file submitted into Teams has saved properly.

Students agree that all work completed in this SAC is their own work and they have not been assisted in the task by another individual.

Students must not disclose the contents of the task; to do so will be a breach of VCE guidelines and will be dealt with according to VCAA regulations.

Instructions

Answer **all** questions in the spaces provided.
To obtain full marks for your responses you should:

- Give simplified answers with an appropriate number of significant figures to all numerical questions; unsimplified answers will not be given full marks.
- Show all working in your answers to numerical questions. No credit will be given for an incorrect answer unless it is accompanied by details of the working.
- Make sure chemical equations are balanced and that the formulas for individual substances include an indication of state; for example, $H_2(g)$; NaCl(s).
- Chemical formulas can be written using regular size numbers (e.g. ethanol will be CH3CH2OH, calcium ion will be Ca2+)

Question 1 (4 marks)

Amino acids are the monomers for proteins.

a. Describe the general structure of an α-amino acid. 1 mark

Answer:

b. Explain what distinguishes an amino acid as essential and non-essential. 1 mark

Answer:

c. Explain the protonation state of all acidic and basic groups in the amino acid leucine at pH 11.0. 2 marks

Answer:

Question 2 (7 marks)

Enzymes are crucial for the reactions involved in the metabolism of food in the human body.

Salivary amylase has an optimum pH of 6.8, whereas in stomach pepsin has an optimum pH of 1.5.

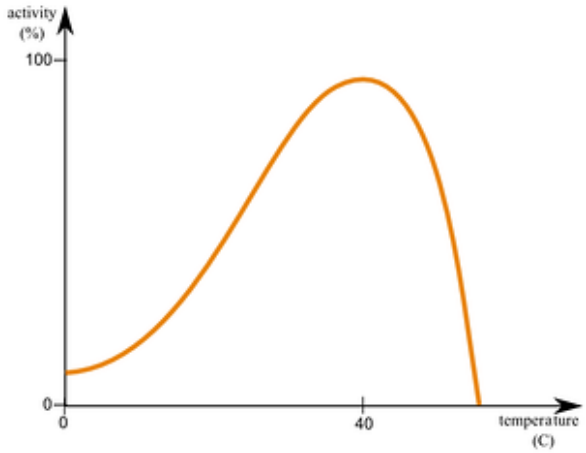
a. Explain why different enzymes (amylase and pepsin) are required to catalyse different functions in the body. 1 mark

Answer:

b. Describe what happens to the amylase produced in the mouth once it reaches the stomach. 2 marks

Answer:

c. The graph below shows the rate of enzyme activity of pepsin in relation to temperature.



Use the graph to explain what happens to the enzyme activity of pepsin:

i. **below** the **optimal** temperature. 2 marks

Answer:

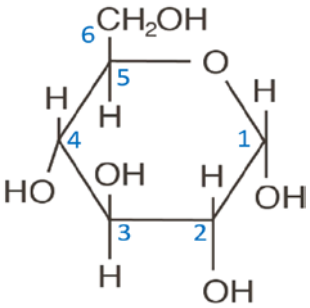
ii. **above** the **optimal** temperature. 2 marks

Answer:

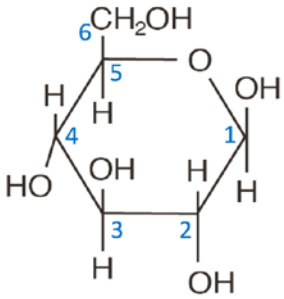
Question 3 (5 marks)

Glucose is one of the body's preferred sources of fuel which is obtained from bread, fruits, vegetables, and dairy products and it exists in two forms.

a. Explain whether the two different glucose molecules shown below are examples of stereoisomers. 2 marks



Glucose molecule **I**



Glucose molecule **II**

Answer:

b. The following table lists a student's summary notes about glucose. It contains some correct and incorrect statements.

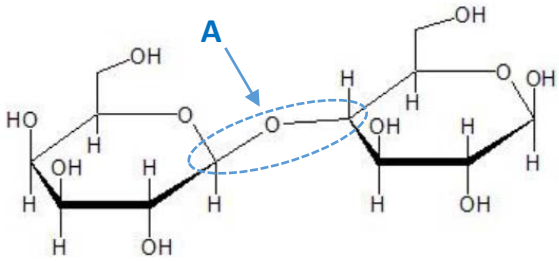
Number	Statement
1	Sucrose is a disaccharide formed by a condensation reaction between the monomers, glucose and fructose.
2	The monosaccharides glucose, galactose and fructose are structural isomers, and they are highly soluble in water.
3	The monosaccharides glucose and galactose both have four chiral centres, but fructose only has three chiral centres.
4	Glucose molecules can polymerise to form two different forms of starch: amylopectin and amylose. Amylose is more soluble than amylopectin.

Identify **one incorrect statement** by writing the statement number in the table provided below. Explain why the statement is incorrect. 3 marks

Statement number	Explanation
Answer:	Answer:

Question 4 (4 marks)

Lactose is a disaccharide which is naturally found in milk and milk products. It is typically broken down by lactase, an enzyme in the small intestine. Some people cannot digest lactose and are known to be lactose intolerant. The structure of lactose is shown below.



a. Name the link labelled as **A** in the lactose molecule shown above. Explain how this link is formed and name the type of reaction. 3 marks

Answer:

b. Name the products formed during the hydrolysis of lactose. 1 mark

Answer:

Question 5 (7 marks)

A popular Australian take away meal is “Fish n’ chips”. The oil used for the deep frying of the food contains the fatty acids: palmitic acid, stearic acid and linoleic acid.

a. Name the common functional group for these fatty acids. 1 mark

Answer:

b. The following table shows the melting points for these fatty acids.

Fatty acid	**Melting point (^{o}C)**
palmitic acid	63
stearic acid	70
linoleic acid	-5

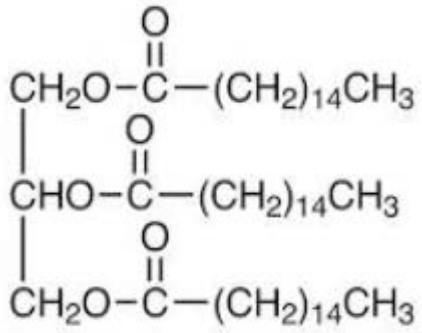
i. With reference to their structure and bonding, explain the difference in melting points between palmitic acid and stearic acid. 2 marks

Answer:

ii. With reference to their structure and bonding, explain the difference in melting points between linoleic acid and stearic acid. 2 marks

Answer:

c. Another commonly used oil for deep frying is palm oil, which mainly consists of glyceryl tripalmitate. The structural formula is shown below:



i. Name the functional group in the glyceryl tripalmitate molecule. 1 mark

Answer:

ii. At the beginning of digestion, glyceryl tripalmitate is broken down into smaller molecules. What is the name given to this general type of chemical reaction? 1 mark

Answer:

END OF QUESTION AND ANSWER BOOKLET